

Project Design Phase-II

Solution Requirements

Date	3rd October 2026
Team ID	6ab8c0aa8c7c66e70673078c
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	

3.2 SOLUTION REQUIREMENTS

The solution requirements define the functional, non-functional, and system requirements needed to implement the proposed access-control mechanism.

3.2.1 FUNCTIONAL REQUIREMENTS

FR1 – User Creation

The system should allow the administrator to create a test user named EEE User with the required user details.

FR2 – Role Creation

The system should support the creation of the following custom roles:

- bb1
- bb2
- bb3
- bb4

FR3 – Role Assignment

The administrator should be able to assign the required roles to the EEE User.

FR4 – Custom Table Creation

The system should contain a custom table:

- Label: Institution Details
- Name: u_institution_details

FR5 – Student Information

The table should contain the following fields:

- Student Roll Number
- Faculty Name
- Email
- Description
- Student Name
- Branch
- Phone Number

FR6 – Branch Values

The Branch field should support:

- ECE
- EEE
- CSE

FR7 – Read Access Control

The READ ACL should control access to records based on the configured role and the Branch condition.

For the defined requirement:

- A user with the bb1 role should be able to view EEE branch records.

FR8 – Create Access Control

The CREATE ACL should require the bb2 role for record creation.

FR9 – Write Access Control

The WRITE ACL should require the bb3 role for modifying records.

FR10 – Delete Access Control

The DELETE ACL should require the bb4 role for deleting records.

FR11 – Unauthorized Access Restriction

Users without the required roles should not be permitted to perform restricted operations.

FR12 – Administrator Verification

The administrator should be able to verify the configured ACL behavior and access the records for administrative purposes.

FR13 – ACL Testing

The system should be tested using different users and combinations of roles to verify the configured access permissions.

3.2.2 NON-FUNCTIONAL REQUIREMENTS

Requirement	Description
Security	The system should prevent unauthorized access to records and operations.
Reliability	ACL rules should consistently enforce the configured permissions.
Maintainability	Roles and ACL configurations should be manageable by the administrator.
Usability	The access-control structure should be clear and understandable.
Performance	ACL evaluation should occur efficiently during record operations.
Scalability	The configuration should be capable of supporting additional users and records.

3.2.3 SYSTEM REQUIREMENTS

Software Requirements

- ServiceNow instance
- Web browser
- ServiceNow administrator account
- security_admin role for ACL configuration
- Custom ServiceNow table
- Test user accounts
- Custom roles
- ServiceNow ACL functionality
- Server-side scripting support

Hardware/Connectivity Requirements

- Computer or laptop
- Stable internet connection
- Compatible web browser

3.2.4 ROLE-BASED ACCESS REQUIREMENTS

Role	Required Operation	Access
bb1	Read	View permitted EEE records
bb2	Create	Create records
bb3	Write	Edit records
bb4	Delete	Delete records

The roles are progressively combined during testing:

